A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 29

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark for the point below. **1** Temperature inversion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Temperature inversion** → a layer of warm air above cooler air that traps pollutants	1
01.2	**1 mark each** 1 mark for each of the 4 points below. **1** On night A, temperature falls with altitude – warm effluent gases are less dense / more buoyant than the surrounding air so they rise **1** As they rise they cool but the surrounding air is also cooler so they stay buoyant and continue to rise, dispersing and diluting **1** On night B the air near the ground is colder / denser than the air above (inversion layer at about 400 m) **1** Pollutants cool as they rise and become denser than the warmer air above so they are trapped below the inversion layer and become concentrated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Cloudless sky allows infrared energy to be radiated from the ground so the ground cools rapidly **1** The cold ground cools the air in contact with it, producing a cold dense layer near the ground **1** Low wind velocity means the air layers of different temperatures do not mix **1** Valley allows cold dense air to collect / drain into the low ground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	3
01.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Smoke smogs (eg London 1952) **1** Photochemical smogs (eg Los Angeles, Mexico City) **1** Build-up of NOx / carbon monoxide / PM10 in cities **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **PM10** → fine solid particles / smoke and dust **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 241-242; AQA spec 3.4.2.2.	
02.1	**1 mark each** 1 mark for each of the 4 points below. **1** Inorganic mercury settled into anaerobic sediments where anaerobic microbes converted it to organic (methyl) mercury **1** Methyl mercury is liposoluble so it is absorbed easily and stored in fat / passes through cell membranes **1** It bioaccumulated in fish and shellfish and biomagnified along the food chain to reach much higher concentrations than in the water **1** Local people ate large amounts of fish (critical group) so received toxic doses **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **critical group** → the people most at risk / most heavily exposed **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Liposoluble so it crosses the blood-brain barrier into the brain, inhibiting enzymes / nerve function (neurotoxin) **1** It crosses the placenta and harms the developing unborn baby **1** Mercury is a teratogen, preventing normal gene expression / development **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** Elemental (liquid) mercury: not easily absorbed through skin or gut, though vapour is absorbed in the lungs **1** Inorganic compounds (eg mercury oxide): moderately well absorbed in the gut **1** Organic compounds (eg methyl mercury): easily absorbed through skin, gut and lungs – liposoluble so most toxic and bioaccumulates **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 276-277; AQA spec 3.4.3.2.11.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 3 points below. **1 Choosing by eye is biased:** people are drawn to patches that look interesting, flowery or easy to reach, so the sample does not represent the whole field **1** Random numbers used as coordinates on a grid give every point an equal chance of being sampled, so the sample is representative and the mean can be applied to the whole area **1** An unbiased sample makes the comparison between the two fields valid, because any difference found is due to the grazing and not to how the sites were chosen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area • **random** → positions chosen using random numbers so the choice is not biased • **valid** → actually measures what it is meant to measure **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	3
03.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** N = (number in first sample x number in second sample) / number marked in second sample = (60 x 75) / 15 **1** = 300 grasshoppers **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • **Lincoln index** → estimating population size from the proportion of marked animals recaptured **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The marked individuals mix randomly back into the population before the second sample is taken **1** The mark does not affect the animal's survival or behaviour, is not lost, and does not make it more visible to predators or to the collector **1 The population is closed:** there are no births, deaths, immigration or emigration between the two samples **1** Marked and unmarked individuals are equally likely to be caught in the second sample **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **valid** → actually measures what it is meant to measure **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	3
03.4	**1 mark each** 1 mark for each of the 3 points below. **1** The quadrat must be large enough to contain a workable number of individuals of the species present but small enough to count accurately - a small quadrat for low-growing grassland plants, a larger one for shrubs **1 A pilot study is used:** the cumulative number of species found is plotted against the number of quadrats, and enough quadrats are taken for the curve to level off **1** More quadrats reduce the effect of chance on the mean and make the result more reliable, but the time available limits how many can be done, so a compromise is reached **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quadrat** → a square frame of known area used to sample a fixed area • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3

Do not write outside the box

Question	Marking guidance	Marks
03.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Survey both fields on the same day, or in the same weather, so that differences in temperature, rainfall or season do not affect the result **1** Use the same method, the same quadrat size and the same number of samples in each field, so the only difference between them is the grazing **1** Choose fields that match in soil type, aspect, slope and altitude, so grazing really is the only variable being tested **1** Identify the plants with the same key, and have the same person make the estimates, so judgements such as percentage cover are consistent **1** Repeat the whole survey and calculate means, and record the spread of the data so the variability within each field can be compared with the difference between them **1** Avoid trampled areas near gates and the field edges, where conditions are untypical of the field as a whole **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **reliable** → repeatable / gives consistent results when repeated • **valid** → actually measures what it is meant to measure • **quadrat** → a square frame of known area used to sample a fixed area **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 394-402; AQA spec 3.7.2.1-3.7.2.2.	
04.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 7×10^{11} tonnes **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (tonnes) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 10% of 10 000 = 1000 billion tonnes **1** 1000 / 700 x 100 = 143% (accept 142.9) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Carbon is transferred into and out of rocks extremely slowly (fossilisation / sedimentation, weathering, volcanic release) **1** Residence time is millions of years so the flux to the atmosphere is tiny compared with respiration / photosynthesis / combustion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
04.4	**1 mark each** 1 mark for each of the 3 points below. **1** CO2 dissolves in the sea (forming carbonic acid / hydrogen carbonate ions) and is absorbed by phytoplankton through photosynthesis **1** Organisms such as molluscs, corals and planktonic organisms build calcium carbonate shells / exoskeletons **1** When they die the shells sink and sediment on the seabed, becoming limestone / chalk over long periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
04.5	**1 mark each** 1 mark for each of the 2 points below. **1** Aerobic respiration uses oxygen to break down organic compounds completely to CO2 (and water), releasing much more energy **1** Without oxygen, anaerobic organisms only partly break down organic compounds, releasing carbon as methane (CH4), which is later oxidised to CO2 in the atmosphere **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-149; AQA spec 3.2.4.2.	
05.1	**1 mark each** 1 mark for each of the 4 points below. **1 Vehicles:** high energy density (200 Wh kg^{-1}) minimises battery mass so range is greater – many cycles and high efficiency **1 Stationary storage:** mass is unimportant so the low energy density of lead-acid does not matter – lowest cost per kWh – well-established, recyclable **1 Lead-acid drawbacks:** fewer cycles and lower efficiency – lithium fire risk needs management **1** Nickel-cadmium largely replaced because of cadmium toxicity and cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
05.2	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Recharging speed **1** Self-discharge during non-use **1** Availability of raw materials (lithium, cobalt) **1** Recyclability / disposal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Parked electric vehicles plugged into the grid form a very large distributed battery **1** At peaks a small proportion of their stored energy is fed back to the grid – surplus grid energy recharges them - avoiding costly, low-efficiency open-cycle gas turbines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surplus electricity electrolyses water to hydrogen, which can be converted to methane **1** Methane is fed into the existing natural gas pipe network and storage, so no new distribution system is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 223-224; AQA spec 3.3.4.1.	
06.1	**1 mark each** 1 mark for each of the 2 points below. **1** The void fills with water naturally – wetlands are uncommon / valuable habitats in most regions **1** The new habitat may have greater wildlife value than the farmland lost – low cost compared with filling and levelling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.2	**1 mark each** 1 mark for each of the 2 points below. **1 Housing:** ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) **1 Farmland:** landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	1 mark for the point below. **1** Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost – transplanted plants often fail **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Aerobic decomposition produces little methane and returns nutrients to the soil	1
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** It may contain concentrated heavy metals / toxic residues **1** It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
07.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Impermeable polymer / clay liner to stop leachate escaping **1** Collection and treatment of leachate **1** Collection of methane and use as fuel **1** Daily covering of waste / final capping and restoration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Impermeable** → does not let water through • **leachate** → polluted liquid draining out of waste **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** C **Also accept** • **stratospheric ozone** → the ozone in the stratosphere that absorbs UV	1
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms **1** Tropospheric ozone is a secondary pollutant (from NOx photochemistry) that is toxic to plants and humans and is a greenhouse gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **Stratospheric ozone** → the ozone in the stratosphere that absorbs UV	2

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Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere **1** So most do not persist long enough to reach the stratosphere where UV would release their chlorine **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CFCs** → CFCs / halogenated organic compounds releasing chlorine **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960) **1** Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (200 - 25) / 200 x 100 **1** = 87.5% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator) **1** Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths **1** Higher doses caused sterility / death **1** After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery) **1** Breeding success recovered so the population rose **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	4
09.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** DDT persists in soil / sediments / food chains for many years so exposure continued **1 Birds of prey are k-selected:** few young per year, slow to mature so populations grow slowly **1** Other pressures (habitat loss, persecution) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
10.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 16.0 - 5.0 = £11 million **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (£ million) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
10.2	**1 mark each** 1 mark for each of the 3 points below. **1** It is rarely practical to reduce emissions to zero and the impact of low-level emissions may be acceptable, so an acceptable level must be decided **1 ALARA:** emissions should be As Low As Reasonably Achievable **1 BATNEEC:** the Best Available Technology Not Entailing Excessive Cost is chosen to achieve this **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

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Question	Marking guidance	Marks
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** As industries expand or cities grow the number of pollutant sources increases **1** So the total emissions / combined concentration rises even if each source is unchanged, and higher standards are needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Critical Pathway Analysis:** predicting movement of mobile persistent pollutants (state of effluent, wind, currents, geology, pH / oxygen and solubility, bioaccumulation, food sources, half-lives) so monitoring focuses on sites at risk – mainly radioactive discharges – checked by sampling • **Critical Group Monitoring:** exposure of the most-exposed group monitored before health effects – if their risk is acceptable everyone else is safer – emissions reduced if not • **Emission control strategies:** location (marine discharges, downwind of cities, not over aquifers) and timing (avoiding temperature inversions, low river flow) • **Principles:** polluter pays (incentive depends on penalties and enforcement), precautionary principle (assume harm until shown otherwise; ignorance no excuse), ALARA / BATNEEC and the non-linear cost-efficiency relationship • **Selection of methods:** production prevention (desulfurisation), prevention of release (precipitators, catalytic converters), post-release remediation (oil spill clean-up, phytoremediation), alternative processes (electric vehicles, renewables, pyrethroids) • **Evaluation:** prevention usually cheaper and more effective than remediation – costs rise steeply near 100% – CPA relies on understanding of pathways – enforcement and international agreement needed for transboundary pollutants – judgement about acceptable risk is political as well as scientific • Justified conclusion	9

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Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **persistent** → does not break down / stays in the environment for a long time • **bioaccumulation** → a pollutant building up inside one organism over time • **Critical Group** → the people most at risk / most heavily exposed **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	
	Model answer - Level 3 (8 marks)	
	Deciding how much to control pollution starts with predicting where a pollutant will go. Critical Pathway Analysis uses the state of the effluent, wind and currents, geology, the effect of pH and oxygen on solubility, bioaccumulation and local food sources to predict whether a discharge will disperse harmlessly or concentrate somewhere sensitive; it is mainly used for radioactive discharges and means monitoring can focus on the sites at risk rather than sampling everywhere. Critical Group Monitoring then checks the exposure of the people most at risk from their lifestyle before any health effects occur: if their risk is acceptable everyone else's is lower, and if not, emissions are reduced.	
	The choice of control is guided by principles. The polluter pays principle gives a financial incentive to prevent damage, but only if penalties are high and polluters cannot escape responsibility. The precautionary principle assumes a waste is harmful until research shows otherwise, so ignorance is no defence. Because the cost of control rises steeply as emissions approach zero, emissions are set As Low As Reasonably Achievable using the Best Available Technology Not Entailing Excessive Cost. Where emission location and timing can be chosen (marine outfalls, downwind of towns, not over aquifers; avoiding temperature inversions) severity is reduced cheaply.	
	Overall, prevention of production (fuel desulfurisation) or of release (electrostatic precipitators, catalytic converters) is more effective than post-release remediation such as oil-spill clean-up, which never recovers all the pollutant. CPA depends on understanding pathways, and transboundary pollutants need international enforcement, but together these principles give a rational basis for deciding acceptable risk rather than aiming for an unaffordable zero.	
	Weaker answer - Level 1 (3 marks)	
	Pollution can be controlled in lots of ways. Critical Pathway Analysis looks at where pollution goes and Critical Group Monitoring looks at people. The polluter pays principle means the person who pollutes has to pay. The precautionary principle means being careful. Pollution can be prevented, or cleaned up afterwards with things like booms for oil spills. Electric cars are an alternative process. Pollution control can be expensive so ALARA is used. All of these are good ways of controlling pollution.	
	What makes the difference	

Do not write outside the box

Question	Marking guidance	Marks
	• The Level 3 answer explains *how* each principle works (why penalties matter, why costs rise near zero emissions); the weak answer only names them.	
	• The Level 3 answer groups the material logically (predict, decide, select) rather than listing everything at the same weight.	
	• 'Evaluate' needs comparison and a judgement: prevention vs remediation, CPA's dependence on understanding pathways, and a conclusion about acceptable risk. The weak answer ends with 'all of these are good'.	
	• Precise examples (desulfurisation, precipitators, catalytic converters, temperature inversions) show applied knowledge; 'things like booms' does not.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	

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Question	Marking guidance	Marks
11.1	**Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility – DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • **Reducing impact:** restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • **Alternatives:** cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • **Trade-offs:** yield, cost, human health (organophosphates), resistance evolution, pollinator services – risk:benefit – sustainability and the circular economy	25
	• **Judgement:** no pesticide is perfectly specific – combination of property-based selection and integrated control is most sustainable *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	

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Question	Marking guidance	Marks
11.2	**To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Fossil fuel features that shaped society: energy density, storage, transport, high temperatures, liquid vehicle fuels, established infrastructure • **Renewable features:** intermittent / unpredictable (wind, solar) vs predictable (tidal) vs controllable (HEP, biofuel, geothermal) – low energy density – location – low pollution but manufacturing impacts • Technologies and developments for each resource with examples – new technologies improving usability • **Storage and grid management:** pumped storage, batteries, hydrogen economy, P2G, V2G, thermal storage, peak shaving – the role of electricity as a secondary fuel • **Economics and politics:** development costs, interest, subsidies, external costs of fossil fuels, falling renewable costs • Energy conservation as part of the solution – sustainability principles - natural systems run on low-density renewable energy • **Judgement:** electricity largely yes	25
	– heat, aviation, shipping and high-temperature industry harder – transition period with nuclear / CCS *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-235; AQA spec 3.3.2-3.3.4.	

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